

electromagnetic field and converts it back into electric current to charge the battery. The two coils, when placed in proximity, form an electrical transformer.

Greater distances between sender and receiver coils can be achieved when the inductive charging system uses resonant inductive coupling. Recent improvements to the resonant system include using a movable transmission coil, for example, mounted on an elevated platform or arm, and the use of silver-plated copper or sometimes aluminium for the receiver coil to minimise weight and decrease resistance due to skin effect.

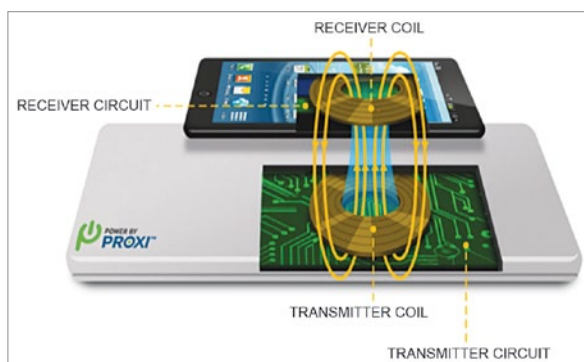
Through this process, power can be transferred safely over an air gap and also through any non-metallic object that might exist between the coils, such as wood, plastic or granite. Addition of extra (or larger) transmitter coils can also extend the range at which power can be transferred. Wireless charging is based on the principle of magnetic resonance or inductive power transfer, the process whereby electricity is transferred between two objects through coils using the following key steps:

1. Mains voltage is converted into high-frequency AC.
2. The AC is sent to the transmitter coil by the transmitter circuit. It then induces a time-varying magnetic field in the transmitter coil.
3. The AC flowing within the transmitter coil induces a magnetic field that extends to the receiver coil (when within a specified distance).
4. The magnetic field generates current within the receiver coil of the device. The process whereby energy is transmitted between the transmitter and the receiver coil is also referred to as magnetic or resonant coupling, and is achieved by both the coils resonating at the same frequency.
5. The AC current flowing

within the receiver coil is converted into DC by the receiver circuit, which can then be used to charge the battery.

Wireless charging technology

2D charging solution. Current options for wireless charging of smartphones and tablets involve the use of a charging pad that acts as the transmitter, transferring power to a miniaturised wireless receiver integrated into smartphones or tablets. This is the safest and most-efficient wireless charging pad technology for electronic devices like smartphones, phablets and tablets, enabling simul-



Resonant wireless charging provides a convenient and safe means to reliably charge and power millions of consumer electronic and industrial devices

taneous wireless charging of multiple devices, with full spatial freedom for placement of devices in any position (no exact alignment).

Miniaturised receivers are integrated directly into the devices, removing the need for external housing covers or sleeves on devices. With continued advances in wireless charging technology, transmitters built into workstations, desks, tables and a whole range of everyday furniture will soon become a reality.

Wireless charging for wearables and devices using AA batteries.

In order to truly deliver wireless charging for all consumer electronic devices, technology needed to somehow enable charging of irregular shaped objects, that is, not only thin form or flat in shape. Such devices commonly use AA batteries or spe-

cialised small batteries in the case of wearable devices. This is called in-device charging and it means convenient, simultaneous charging of multiple devices including remote controllers, gaming devices, cameras, toys, wearable devices and more, without having to remove and replace batteries.

Wireless ambient power. Companies are developing a technology that could draw enough power from ambient radio waves to keep a mobile phone handset topped up. Ambient electromagnetic radiation—emitted from Wi-Fi transmitters, mobile phone antennae, TV masts and other sources—could be converted into enough electrical current to keep a battery topped up. They are working towards a prototype that could harvest up to 50 milliwatts of power—enough to slowly recharge a phone that is switched off. Current prototypes can harvest three to five milliwatts.

Wirelessly powered light bulb. Researchers have created a revolutionary device that could remotely charge batteries and power household appliances. They have shown that it is possible to wirelessly power a 60-watt light-bulb sitting about two metres away from a power source. They have demonstrated, for the first time, that it is feasible to efficiently send that much power over such distance. The experiment paves the way for wirelessly charging batteries in laptops, mobile phones and music players, as well as disposing of electric cords on household appliances.

Benefits

Whatever the application, removal of the physical connection delivers a number of benefits over traditional cable connections, some of which are not always obvious. This article highlights below some of the benefits and advantages of reso-